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United States Patent	12397593
Kind Code	B2
Date of Patent	August 26, 2025
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Sound-absorbing device for a vehicle wheel

Abstract

A sound-absorbing device includes a resonator. The resonator includes a flexible, tube-shaped resonator body having a first end, a second end, and a channel fluidically connecting the first end and the second end. The resonator body is wrapped completely around a hub of a vehicle wheel at least one time and is configured to absorb sound produced within a cavity of a tire of the vehicle wheel.

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Appl. No.: 18/112754

Filed: February 22, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240286440 A1	Aug. 29, 2024

Publication Classification

Int. Cl.: B60C19/00 (20060101); B60B21/02 (20060101)

U.S. Cl.:

CPC B60C19/002 (20130101); B60B21/026 (20130101); B60B2900/133 (20130101)

Field of Classification Search

CPC: B60B (2900/133); B60B (21/026); B60C (19/002)

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to sound absorbing devices and, more particularly, to sound absorbing devices for vehicle wheels.

BACKGROUND

(2) Vehicle wheels typically include a hub and a tire mounted to the hub. The tire is usually hollow

and includes a cavity, and in some instances, sound is produced in the cavity as the tire rolls on the ground during driving of the vehicle. This sound can be undesirable to a driver of the vehicle. The present disclosure addresses issues related to mitigating sound produced by vehicle wheels.

SUMMARY

(3) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(4) In one form of the present disclosure, a sound-mitigating device includes a resonator. The resonator is configured to mitigate sound within a cavity of a tire of a vehicle wheel. The resonator includes a flexible, tube-shaped resonator body having a first end, a second end, and a channel fluidically connecting the first end and the second end. The resonator body is configured to wrap completely around a hub of a vehicle wheel at least one time and absorb sound produced within the cavity.

(5) In another form of the present disclosure, a vehicle wheel includes a hub and a tire mounted to the hub and defining a hollow cavity. The vehicle wheel also includes a resonator. The resonator is configured to mitigate sound within the cavity. The resonator includes a flexible, tube-shaped resonator body having a first end, a second end, and a channel fluidically connecting the first end and the second end. The resonator body is configured to wrap completely around the hub at least one time. The resonator is configured as a quarter-wavelength resonator having a resonance frequency that is a function of a length of the resonator body and that is substantially equal to a resonance frequency of a sound produced by the tire rolling on the ground.

(6) Further areas of applicability and various methods of enhancing the above technology will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present teachings will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. 1A shows an example of a vehicle wheel including a hub and a tire.

(3) FIG. 1B shows an example of a cross-section of the vehicle wheel including a sound-absorbing device having a resonator.

(4) FIG. 2 shows an example of the resonator filled with sound-absorbing material.

(5) FIG. 3 shows an example of a mount for adhering the resonator to the hub.

(6) FIG. 4 shows an example of an end cap for the resonator.

(7) FIG. 5A shows an example of the resonator sectioned into resonator body sections using clamps.

(8) FIG. 5B shows a cross-sectional view of a portion of the resonator body including a clamp.

(9) FIG. 5C shows a cross-sectional view of a portion of the resonator sectioned into resonator body sections using a bead.

(10) FIG. 6A shows an example of multiple resonators for the vehicle wheel.

(11) FIG. 6B shows an example of the resonator wrapped multiple times around the hub.

(12) FIG. 7A shows an example of an acoustic spectra of a tire without a resonator.

(13) FIG. 7B shows an example of an acoustic spectra of a tire with a resonator having an inner diameter of 10 (millimeters) mm.

(14) FIG. 7C shows an example of an acoustic spectra of a tire with a resonator having an inner diameter of 15 mm.

(15) FIG. 7D shows an example of multiple resonance modes of a resonator having a longer length.

DETAILED DESCRIPTION

(16) The present disclosure provides a sound-absorbing device for a vehicle wheel. The sound-absorbing device includes a resonator. The resonator includes a flexible, tube-shaped resonator body having a first end, a second end, and a channel fluidically connecting the first end and the second end. The resonator body is wrapped completely around a hub of a vehicle wheel at least one time and is configured to absorb sound produced within a cavity of a tire of the vehicle wheel.

(17) Referring to FIGS. 1A and 1B, an example of a wheel **100** for a vehicle is shown. Among other components typical of vehicle wheels, the wheel **100** includes a tire **110** and a hub **120**. The tire **110** can be a standard tire typical of vehicles and formed from rubber. The tire **110** is hollow, defines a cavity **130**, and is configured to be mounted to the hub **120**. As the wheel **100** rolls on a ground upon which the vehicle is traveling, sound may be produced within the cavity **130**. In some instances, this sound can be undesirable to a driver of the vehicle. Therefore, it may be advantageous to absorb at least some of the sound produced within the cavity **130**.

(18) Accordingly, the wheel **100** includes a sound-absorbing device. The sound-absorbing device is located within the cavity **130** and is configured to absorb sound produced within the cavity **130** as the wheel **100** rolls on the ground. The sound-absorbing device includes a resonator **140**. The resonator **140** includes a resonator body **150**, which is long, flexible, and tube-like and can be configured to wrap around the hub **120** one or more times. As used herein, the term “flexible” means that the resonator body **150** is pliable enough to wrap around the hub **120** without incurring significant stresses in the material of the resonator body **150** that might lead to the material ripping or breaking. The resonator body **150** also includes a first end **160** and a second end **170** and defines a channel **180** (FIG. 2) fluidically connecting the first end **160** and the second end **170**. As molecules of air or other gas within the cavity **130** pass through the channel **180**, the channel **180** significantly restricts movement of the molecules, and friction between the molecules increases. Accordingly, acoustic energy is dissipated in the form of heat due to the friction, and sound is absorbed by the resonator **140**.

(19) As described herein, the channel **180** is substantially hollow. However, in one or more arrangements, referring to FIG. 2, the channel **180** may be filled with sound-absorbing material **200**. As shown, the sound-absorbing material **200** fills the entire channel **180**. However, it will be understood that, in some arrangements, the sound-absorbing material **200** does not fill the entire channel **180**. For example, only a section of the channel **180** may be filled with sound-absorbing material **200**. In another example, the entire channel **180** may be lined with sound-absorbing material **200**, or only a section of the channel **180** may be lined with sound-absorbing material **200**.

(20) Referring back to FIGS. 1A and 1B, in one or more arrangements, the resonator body **150** is formed from flexible tubing. For example, the resonator body **150** can be formed from commercially-available silicone or rubber tubing. However, any other suitable material may be used to form the resonator body **150**. In some instances, the material forming the resonator body **150** is lightweight. The resonator body **150** can be formed in any suitable manner. For example, the resonator body **150** can be formed by cutting commercially available tubing to a desired size. In other examples, the resonator body **150** can be formed by extruding, molding, or otherwise forming plastic or silicone-based material into a tube shape. Once formed to the desired size, the resonator body **150** defines an inner diameter D and a length L (FIG. 3). The inner diameter D and/or the length L can be any suitable dimension. In some instances, the inner diameter D has the same dimension along the length L of the resonator body **150**. In other instances, the inner diameter D may vary along the length L of the resonator body **150**. The length L , in some instances, is substantially equal to or greater than a circumference of the hub **120**.

(21) The resonator body **150** can be attached to the hub **120** in any suitable manner. In one or more arrangements, the resonator body **150** is adhered directly to the hub **120**. For example, the resonator body **150** can be glued or taped directly to the hub **120**. In one or more other arrangements, referring now to FIG. 3, the resonator body **150** can be attached to the hub **120** using

a mount **300**. The mount **300** can be unitarily formed with the hub **120**, or the mount **300** can be configured for attachment to the hub **120**. The mount **300** can include one or more apertures configured to hold the resonator body **150** at a location near both the first end **160** and the second end **170**, as shown. In one or more arrangements, the aperture(s) can have a diameter slightly larger than an outer diameter of the resonator body **150**, and the resonator body **150** can be adhered to the mount **300**, for example, by glue. In one or more other arrangements, the aperture(s) can have a diameter substantially equal or less than the outer diameter of the resonator body **150** such that the resonator body **150** forms a tight fit (e.g., a press fit) with the aperture(s) to keep the resonator **140** in place.

(22) In one or more arrangements, the resonator **140** is a quarter-wavelength resonator **140**. As mentioned above, the resonator body **150** defines an inner diameter D and a length L. In some instances, the inner diameter D of the resonator body **150** governs the bandwidth of the sound absorbed by the resonator **140**. Accordingly, the inner diameter D can have any suitable dimension that creates a high friction between the air molecules in the channel **180**. Moreover, in some instances, the length L of the resonator body governs the resonance frequency f of the resonator **140**. For example, the resonance frequency f is defined by the following equation, where c is the speed of sound and n is the harmonic index (n=1, 3, 5, 7, etc.). The harmonic index will be described in further detail below.

$$(23) f, n = \frac{cn}{4L}$$

(24) In some instances, it may be advantageous to tune the resonance frequency f to a frequency of the sound produced by the tire **110** rolling on the ground as the vehicle is driven, which, in some instances, is about 500-1000 Hz. However, it will be understood that the resonance frequency f may be tuned to any other suitable resonance frequency. In one or more arrangements, the resonance frequency f can be changed by the addition of one or more components to the resonator **140**. For example, referring now to FIG. 4, the resonator **140** may include a cap **400**. The cap **400** can be configured to change the resonance frequency f, for example, by changing the speed of the sound traveling through the channel **180**. More specifically, a resonator **140** with two open ends may function as a different type of resonator rather than a quarter-wavelength resonator. The cap **400** can be attached to the first end **160** or the second end **170**, and the cap **400** can be attached to the resonator **140** in any suitable manner. For example, the cap **400** can define an aperture configured to receive the first end **160** or the second end **170**. In one or more arrangements, the aperture can have a diameter slightly smaller than an outer diameter of the resonator body **150**, and the resonator body **150** can be adhered to the cap **400**, for example, by glue. In one or more other arrangements, the aperture can have a diameter substantially equal to or less than the outer diameter of the resonator body **150** such that the resonator body **150** forms a tight fit (e.g., press fit) with the aperture(s) to keep the cap **400** in place.

(25) Though the tire **110** is described above as exhibiting a single resonance frequency, in some instances, the tire **110** may exhibit multiple resonance frequencies. Accordingly, it may be advantageous to design the resonator **140** such that it is configured to absorb sound waves of different frequencies. In one or more arrangements, referring now to FIG. 5A, this may be accomplished by sectioning the channel **180** (FIG. 2) to create two or more resonator body sections **500**. Each resonator body section **500** defines its own aperture **510**, which allows the passage of air or other gas through the channel **180** of each resonator body section **500**, thereby allowing each resonator body section **500** to function as an individual resonator.

(26) As shown in FIG. 5A, the channel **180** is sectioned to form five (5) resonator body sections **500**. The five resonator body sections include a first resonator body section **500A**, a second resonator body section **500B**, a third resonator body section **500C**, a fourth resonator body section **500D**, and a fifth resonator body section **500E**. However, it will be understood that the channel **180** can be sectioned to form any suitable number of resonator body sections **500**, for example, two (2) resonators body sections **500**, three (3) resonator body sections **500**, four (4) resonator body

sections **500**, six (6) resonator body sections **500**, etc. As mentioned above, each resonator body section **500** defines its own aperture **510**. More specifically, the first resonator body section **500A** defines a first aperture **510A**, the second resonator body section **500B** defines a second aperture **510B**, the third resonator body section **500C** defines a third aperture **510C**, the fourth resonator body section **500D** defines a fourth aperture **510D**, and the fifth resonator body section **500E** defines a fifth aperture **510E**. In some instances, the first aperture **510A** can be the first end **160**, and the fifth aperture **510E** can be the second end **170**.

(27) The channel **180** can be sectioned in any suitable manner. Referring now to FIGS. 5A and 5B, in one or more arrangements, the resonator **140** can include one or more clamps **520**. The clamp(s) **520** are configured to clamp the resonator body **150** to block the passage of air or other gas at the point of the clamp **520**. As shown in FIG. 5A, the resonator **140** includes four (4) clamps **520** configured to section the resonator body **150** into the five resonator body sections **500**. More specifically, the resonator **140** includes a first clamp **520A**, a second clamp **520B**, a third clamp **520C**, and a fourth clamp **520D**. However, it will be understood that the resonator **140** can include any suitable number of clamp(s) **520** configured to section the resonator body **150** into any suitable number of resonator body sections **500**. The clamp(s) **520** can be positioned along the length **L** of the resonator body **150** in any suitable manner. For example, as shown in FIG. 5A, the clamp(s) **520** are substantially equally spaced along the length **L**. However, it will be understood that the clamp(s) **520** can be un-equally spaced along the length **L**.

(28) The clamp(s) **520** can have any suitable structure. Referring now to FIG. 5B, a cross-sectional view of one arrangement of one of the clamp(s) **520** is shown. The clamp **520** has a clamp body **530** that is configured to close around the resonator body **150**. The clamp body **530** defines a first protrusion **540A** and a second protrusion **540B** respectively positioned across from each other with respect to the resonator body **150**. When the clamp body **530** is clamped to the resonator body **150**, the first protrusion **540A** and the second protrusion **540B** pinch the resonator body **150** such that the passage of air or other gas through the channel **180** at the point of the clamp **520** is substantially blocked.

(29) In one or more other arrangements, referring now to FIG. 5C, the resonator **140** can include one or more beads **550** located in the channel **180** and configured to section the channel **180** into the resonator body sections **500**. The bead(s) **550** can be formed from acoustically-rigid material. As used herein, the term “acoustically-rigid” means that the material significantly blocks sound from passing through the material. The acoustically-rigid material can be a physically rigid material such as plastic or metal, or the acoustically-rigid material can be a physically softer but still acoustically-rigid material such as foam or even drops of water. Like the clamp(s) **520**, the bead(s) **550** can be substantially equally spaced along the length **L** of the resonator body **150**. However, it will be understood that the bead(s) **550** can be un-equally spaced along the length **L**.

(30) Another way to account for multiple resonance frequencies produced by the tire **110** is to use multiple resonators. Referring now to FIG. 6A, the sound-absorbing device can include two resonators. The sound-absorbing device can include a first resonator **140**, as described above. The sound-absorbing device can also include a second resonator **600**. Like the first resonator **140**, the second resonator **600** includes a second resonator body **610**, a second resonator first end **620**, a second resonator second end **630**, and a second resonator channel (not shown) fluidically connecting the second resonator first end **620** and the second resonator second end **630**. Both the first resonator **140** and the second resonator **600** can be configured to wrap around the hub **120** one or more times. In some instances, the first resonator **140** and the second resonator **600** can define different inner diameters and/or different lengths such that the first resonator **140** and the second resonator **600** exhibit different resonance frequencies. For example, the first resonator **140** can define a first inner diameter $D_{sub.1}$ and a first length $L_{sub.1}$, and the second resonator **600** can define a second inner diameter $D_{sub.2}$ and a second length $L_{sub.2}$. In some instances, the first inner diameter $D_{sub.1}$ and the second inner diameter $D_{sub.2}$ are different. Similarly, in some

instances, the first length $L_{\text{sub.1}}$ and the second length $L_{\text{sub.2}}$ are different. However, it will be understood that the first inner diameter $D_{\text{sub.1}}$ and the second inner diameter $D_{\text{sub.2}}$ can be substantially equal, and the first length $L_{\text{sub.1}}$ and the second length $L_{\text{sub.2}}$ can be substantially equal. Moreover, it will be understood that while FIG. 6A depicts the sound-absorbing device as having two resonators, the sound-absorbing device can include any suitable number of resonators. For example, the sound-absorbing device can include 3 resonators, 4 resonators, 5 resonators, etc. (31) Yet another way to account for multiple resonance frequencies produced by the tire **110** is to design the resonator **140** with a longer length L and wrap the resonator **140** around the hub **120** multiple times. As used herein, the term “longer length” can mean that the resonator **140** has a length L that is long enough to wrap around the hub **120** two or more times (in other words, the length L is greater than two times the circumference of the hub **120**). Referring to FIG. 6B, an example of a resonator **140** having a longer length L and being wrapped multiple times around the hub **120** is shown. In arrangements where the resonator **140** is a quarter-wave length resonator **140**, a characteristic of the quarter-wavelength resonator **140** is its ability to have multiple resonance modes for a longer length L . Because the frequency interval is inversely proportional to the length L , the longer the resonator **140** is, the smaller the frequency intervals become such that many resonance peaks appear in a given frequency range, and the more resonance frequencies are observed by the resonator **140**. For example, the fundamental resonance mode (e.g., the fundamental frequency) is $f_{\text{sub.0}}$. Higher-mode resonances occur in increasing intervals of $2 \cdot f_{\text{sub.0}}$. For example, higher-mode resonances occur at $3 \cdot f_{\text{sub.0}}$, $5 \cdot f_{\text{sub.0}}$, $7 \cdot f_{\text{sub.0}}$, etc. Since the higher-mode resonances are multiples of the fundamental resonance, in some instances, it may be advantageous to design the resonator to have the lowest possible fundamental resonance mode. This allows the resonator to have many more higher-mode resonance frequencies for the same length of the resonator body.

(32) Referring now to FIGS. 7A-7C, examples of the acoustic spectra of some of the variations described herein are shown. FIG. 7A shows an acoustic spectrum of the sound produced within a cavity **130** of a tire **110** without a resonator **140**. It can be seen that the sound exhibits multiple resonance frequencies. More specifically, the resonance frequencies occur at around 200 Hertz (Hz), around 370 Hz, around 550 Hz, around 750 Hz, and around 950 Hz. The noise level of the first resonance frequency (about 200 Hz) is about 35 decibels (dB). The noise level of the second resonance frequency (about 370 Hz) is about 90 dB. The noise level of the third resonance frequency (about 550 Hz) is about 65 Hz. The noise level of the fourth resonance frequency (about 750 Hz) is about 105 Hz. The noise level of the fifth resonance frequency (around 950 Hz) is about 80 dB.

(33) FIG. 7B shows an example of an acoustic spectrum of the sound produced within the cavity **130** of the tire **110** including a resonator **140** with an inner diameter of about 10 mm. It can be seen that the noise levels of some of the resonance frequencies described above in connection with FIG. 7A are lower. For example, the noise level of the second resonance frequency (about 370 Hz) is about 80 dB. The noise level of the fourth resonance frequency (about 750 Hz) is about 80 Hz. The noise level of the fifth resonance frequency (around 950 Hz) is about 70 dB.

(34) FIG. 7C shows an example of an acoustic spectrum of the sound produced within the cavity **130** of the tire **110** including a resonator **140** with an inner diameter of about 15 mm. It can be seen that the noise levels of some of the resonance frequencies described above in connection with FIGS. 7A and 7B are even lower. For example, the noise level of the second resonance frequency (about 370 Hz) is about 70 dB. The noise level of the third resonance frequency (about 550 Hz) is about 55 Hz. The noise level of the fourth resonance frequency (about 750 Hz) is about 60 Hz. The noise level of the fifth resonance frequency (around 950 Hz) is about 60 dB.

(35) FIG. 7D shows an example of an acoustic spectrum of the sound produced within a cavity **130** of a tire **110** with a resonator **140** having a longer length, for example, the resonator of FIG. 6B. It can be seen that the resonator **140** exhibits multiple resonance modes, for example, a first

resonance mode at about 1 Hz, a second resonance mode at about 200 Hz, a third resonance mode at about 280 Hz, a fourth resonance mode at about 480 Hz, a fifth resonance mode at about 670 Hz, and a sixth resonance mode at about 870 Hz. These resonance modes may correspond to the resonance frequencies of the sound produced within the cavity **130** as the tire **110** rolls on the ground during driving.

(36) The variations described herein have the benefit of absorbing sound produced within a cavity of a tire of a vehicle wheel as the wheel rolls on the ground during driving. Moreover, the variations described herein allow for sound-absorption within a tire without the need for high-cost manufacturing of sound-absorbing devices within the hub of the wheel itself. Additionally, the variations described herein may allow for low-cost sound absorption, as the resonator can be produced with standard, low-cost materials.

(37) The preceding description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. Work of the presently named inventors, to the extent it may be described in the background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present technology.

(38) As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A or B or C), using a non-exclusive logical “or.” It should be understood that the various steps within a method may be executed in different order without altering the principles of the present disclosure. Disclosure of ranges includes disclosure of all ranges and subdivided ranges within the entire range.

(39) The headings (such as “Background” and “Summary”) and sub-headings used herein are intended only for general organization of topics within the present disclosure and are not intended to limit the disclosure of the technology or any aspect thereof. The recitation of multiple variations or forms having stated features is not intended to exclude other variations or forms having additional features, or other variations or forms incorporating different combinations of the stated features.

(40) As used herein the term “about” when related to numerical values herein refers to known commercial and/or experimental measurement variations or tolerances for the referenced quantity. In some variations, such known commercial and/or experimental measurement tolerances are $\pm 10\%$ of the measured value, while in other variations such known commercial and/or experimental measurement tolerances are $\pm 5\%$ of the measured value, while in still other variations such known commercial and/or experimental measurement tolerances are $\pm 2.5\%$ of the measured value. And in at least one variation, such known commercial and/or experimental measurement tolerances are $\pm 1\%$ of the measured value.

(41) The terms “a” and “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The phrase “at least one of . . . and . . .” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B, and C” includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC, or ABC).

(42) As used herein, the terms “comprise” and “include” and their variants are intended to be non-limiting, such that recitation of items in succession or a list is not to the exclusion of other like items that may also be useful in the devices and methods of this technology. Similarly, the terms “can” and “may” and their variants are intended to be non-limiting, such that recitation that a form or variation can or may comprise certain elements or features does not exclude other forms or variations of the present technology that do not contain those elements or features.

(43) The broad teachings of the present disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure

should not be so limited since other modifications will become apparent to the skilled practitioner upon a study of the specification and the following claims. Reference herein to one variation, or various variations, means that a particular feature, structure, or characteristic described in connection with a form, a variation, or a particular system is included in at least one variation or form. The appearances of the phrase “in one variation” (or variations thereof) are not necessarily referring to the same variation or form. It should be also understood that the various method steps discussed herein do not have to be carried out in the same order as depicted, and not each method step is required in each variation or form.

(44) The foregoing description of the forms and variations has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular form or variation are generally not limited to that particular form or variation, but, where applicable, are interchangeable and can be used in a selected form or variation, even if not specifically shown or described. The same may also be varied in many ways. Such variations should not be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A sound-absorbing device, comprising: a resonator including a flexible, tube-shaped resonator body formed from rubber or silicone tubing having a first end, a second end, and a channel fluidically connecting the first end and the second end, the resonator body wrapped completely around a hub of a vehicle wheel at least one time and configured to absorb sound produced within a cavity of a tire of the vehicle wheel; and one or more beads located in the channel and configured to section the channel into a plurality of resonator body sections.
2. The sound-absorbing device of claim 1, wherein the resonator body is adhered directly to the hub.
3. The sound-absorbing device of claim 1, further comprising a mount configured to adhere the resonator body to the hub.
4. The sound-absorbing device of claim 1, wherein the resonator is a quarter-wavelength resonator having a resonance frequency that is a function of a length of the resonator body.
5. The sound-absorbing device of claim 4, wherein the resonance frequency is substantially equal to a resonance frequency of a sound produced by the tire rolling on a ground surface.
6. The sound-absorbing device of claim 4, further comprising a cap covering one of the first end and the second end, thereby changing the resonance frequency.
7. The sound-absorbing device of claim 1, wherein the one or more beads are acoustically rigid foam beads, and wherein each of the plurality of resonator body sections defines an aperture such that each of the plurality of resonator body sections functions as an individual resonator.
8. A sound-absorbing device comprising: a resonator including a flexible, tube-shaped resonator body formed from rubber or silicone tubing having a first end, a second end, and a channel fluidically connecting the first end and the second end, the resonator body wrapped completely around a hub of a vehicle wheel at least one time and configured to absorb sound produced within a cavity of a tire of the vehicle wheel; and one or more clamps located external to the resonator body and configured to clamp the resonator body to section the channel into a plurality of resonator body sections, such that the plurality of resonator body sections define unequal lengths.
9. The sound-absorbing device of claim 1, wherein the resonator includes sound-absorbing material located in the channel to further absorb sound within the cavity.
10. The sound-absorbing device of claim 1, wherein the resonator body has a length substantially longer than a circumference of the hub, is wrapped completely around the hub at least two times, and is configured to absorb sound of multiple resonance frequencies, the multiple resonance frequencies including a fundamental frequency and one or more higher-mode resonance

frequencies in intervals of twice the fundamental frequency.

11. A vehicle wheel, comprising: a hub; a tire mounted to the hub and defining a hollow cavity; a resonator including a flexible, tube-shaped resonator body formed from rubber or silicone tubing having a first end, a second end, and a channel fluidically connecting the first end and the second end, the resonator body wrapped completely around the hub at least one time, the resonator configured as a quarter-wavelength resonator having a resonance frequency that is a function of a length of the resonator body and that is substantially equal to a resonance frequency of a sound produced by the tire rolling on a ground surface, whereby the resonator is configured to absorb sound produced in the cavity; and one or more beads located in the channel configured to section the channel into a plurality of resonator body sections.

12. The vehicle wheel of claim 11, further comprising a cap covering one of the first end and the second end, thereby changing the resonance frequency.

13. The vehicle wheel of claim 11, wherein the one or more beads are acoustically rigid foam beads, and wherein each of the plurality of resonator body sections defines an aperture such that each of the plurality of resonator body sections functions as an individual resonator.

14. The vehicle wheel of claim 11, wherein the resonator includes sound-absorbing material located in the channel to further absorb sound within the cavity.

15. The vehicle wheel of claim 11, wherein the resonator body has a length substantially longer than a circumference of the hub, is wrapped completely around the hub at least two times, and is configured to absorb sound of multiple resonance frequencies, the multiple resonance frequencies including a fundamental frequency and one or more higher-mode resonance frequencies in intervals of twice the fundamental frequency.
